

Nupur Pushkar Bhatkhande

Final Year B.Tech, AI & Data Science | SPPU Pune, India

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Professional Summary

- Final-year B.E. student (AI & Data Science, CGPA 9.24) with hands-on experience in machine learning, computer vision, and NLP. Built scalable AI systems recognized at international hackathons (Hong Kong, PAN IIT). Passionate about developing responsible, data-driven, and impactful AI solutions in healthcare, fitness, and automation.

Education

2022-2026

- Savitribai Phule Pune University Bachelor of Engineering Artificial Intelligence and Data Science, CGPA: 9.24
- Relevant Coursework:**
Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management, Artificial Intelligence, Machine Learning, Cloud Computing.

Professional Experience

- Python Intern | Core Sphere Technology, India Dec 2023 – Feb 2024**
Collaborated on backend modules using Python and Flask; refactored algorithms to optimize runtime by 25%, improving backend efficiency and maintainability.
- AI/ML Intern | AICTE Eduskills – Google Virtual Internship Jul 2024 – Sept 2024**
Deployed ML models using TensorFlow on Google Cloud, improving image classification accuracy by 15% and enabling large-scale sentiment analysis across 10k+ text samples.
- Data Science Intern | Suvidha Foundation Dec 2024 – Jan 2025**
Conducting NLP research using Hugging Face transformers (PEGASUS) for abstractive text summarization; optimizing preprocessing pipelines for tokenization and fine-tuning transformer-based models.

Publication

Patent: AI-based Smart Fitness Solution (Fitmantra)

- An AI based Smart Fitness Solution (Fitmantra) Application No.: 202421004780 | Published: Feb 2024 Patent granted for a computer vision based fitness system enabling real-time posture correction, gamified motivation, and personalized health planning. Designed to democratize access to quality fitness training through ethical AI, user privacy, and data-driven personalization.

Hackathons Recognition

- Easy-A Hackathon 2025 – Finalist (Hong Kong)**

Winner of 12 blockchain + AI challenges in 24 hours and Finalist among 1200 overseas participants.

- **Navomnesh '24 – 4th Place (Automation)**

Built an innovative automation solution, recognized among 5,000+ participants.

- **Infotsav '24 – 3rd Place (Cybersecurity)**

Developed an award-winning cybersecurity project, ranked among 1,000+ participants. •

PAN IIT Hackathon – 5th Rank

Secured ₹50,000 prize for a novel AI model in a 24-hour challenge.

- **PICT International Hackathon (Healthcare) –**

1st Place Won ₹30,000 for an AI-driven health monitoring system judged by global panel.

- **State-Level MIT Hackathon – 2nd Place | Smart India Hackathon – 5th Place**

Developed time-critical AI solutions, praised for rapid prototyping and interdisciplinary thinking.

Projects

- **Driver's Drowsiness Detection System**

Developed an intelligent eyewear solution using OpenCV and Embedded Vision Systems to monitor eye blinks and alert drivers in case of fatigue (>3 sec eye closure).

Achieved 92% accuracy and real-time response under various lighting conditions. Tech:

Python, OpenCV, Haar Cascade, Eye-Tracking.

- **Skin Cancer Detection using CNN-ANN Hybrid Model**

Implemented a diagnostic model combining CNN for feature extraction and ANN for classification, improving lesion identification accuracy to 94% on dermoscopic datasets. Experienced with Machine learning algorithms.

Tech: TensorFlow, Keras, Image Processing, CNN, ANN.

- **Virtual Fitness Assistant**

Built a real-time AI trainer using OpenCV and MediaPipe Library for posture correction and exercise tracking. Integrated with speech and motion feedback for engagement. Worked with Open CV and other python libraries.

Tech: Python, MediaPipe, OpenCV, Computer Vision.

- **Genome Data Vault (Blockchain + AI for Genomic Research)**

Developing a blockchain-based platform enabling secure, privacy-preserving analysis of genomic data using federated learning and ZKPs for data monetization ethics. Tech: Python, Flask, TensorFlow, Solidity, Metadata, etc.

Certification and Training

- **Google for Developers – AI/ML Virtual Internship (2024)**

- **Tableau for Data Analytics | R for Data Science • AICTE & SPU National Hackathon 2k24 | GDSC Sprint Hackathon 2024**

- **MMCOE Pune: Entrepreneurship & National Technical Events.**

- **Udemy : C++ Programming, Python.**

- **IIT Bombay : National Tech Workshop on Role of Data Science in Sports.**

Technical Skills

- **Programming:** Python, C++, Java, SQL
- **Frameworks:** TensorFlow, PyTorch, Scikit-learn, Hugging Face, OpenCV, Flask
- **Core Areas:** Machine Learning, Deep Learning, NLP, Computer Vision, AI Ethics
- **Cloud & Big Data:** Google Cloud (Vertex AI, BigQuery), AWS (Lambda, SageMaker)
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Dev Tools:** Git, Docker, Jupyter, Tableau
- **Soft Skills:** Leadership, Communication, Research Writing, Team Collaboration.

Awards & Recognition

- **Academic Excellence (Ensemble 2024–25)** – Special performer in AI & DS Department.
- **Patent & Hackathon Recognition** across 6 national, 2 international Hackathon winner and Finalist.
- **Tech Speaker: Presented on AI Ethics & Responsible Deployment (2024-25)**

Interests

Artificial Intelligence Research | Cloud Computing | Responsible AI | Open-Source Contribution | Hackathons & Innovation Challenges